

MARK CUSTARD

SDET – TEST AUTOMATION & FULL-STACK ENGINEERING

Ridgefield, WA · Remote
+1 360 771 0564
mark.a.custard@gmail.com
LinkedIn · GitHub
markcustard.com

SDET and full-stack engineer with 7+ years in quality engineering across fintech, insurance and entertainment platforms. Builds and runs a live booking platform end to end at TAG Parking (React, FastAPI, PostgreSQL, Stripe), backed by a 20,100+ test suite written test-first, with CI gating every push and every production promotion. Built a Playwright framework from zero to 700+ tests for a Netflix acquisition platform, cutting manual regression by 90%. Strongest in test architecture, API and integration testing, payment failure modes and root-cause analysis, and in building with the Claude API.

CORE COMPETENCIES

TEST-FIRST · AUTOMATION-LED · RELEASE-ACCOUNTABLE

01	TESTING & QA	Integration Testing, Regression Testing, Test Automation, Defect Management, Playwright, Selenium, Pytest, Vitest, Manual Testing, A/B Testing, Exploratory Testing
02	API & DATA	API Testing, Postman, SoapUI, FastAPI, Claude API, SQL, PostgreSQL, Cosmos DB, Google BigQuery, Python, Pandas
03	CRM & PLATFORMS	Salesforce, Stripe, Data Extensions, Data Synchronization, Data Validation
04	METHODOLOGIES & TOOLS	Azure DevOps, CI/CD, GitHub Actions, Jira, Agile/Scrum, Business Process Automation, Zapier, React

EXPERIENCE

Full Stack Developer – TAG Parking

Sep 2025 – Present · Remote

- Architect and deliver a complete booking platform (React, FastAPI, PostgreSQL) with complex data synchronization, processing **\$256k** in gross bookings (**\$223k** net revenue across 2,100+ bookings) through automated financial workflows
- Lead full-stack development and business process automation by implementing a comprehensive auto-roster scheduling engine, including the fix for a defect that silently undid staff edits whenever a new booking arrived (proven by 28 injected faults and two independent adversarial reviews)
- Build and maintain a **20,100+ test suite** (14,400 pytest, 4,500 Vitest, 1,200 Playwright E2E) written test-first, with CI gating every push and every staging-to-production promotion – including “census” tests that enumerate every code path of a kind and prove each check by injecting a fault
- Integrate robust Stripe payment architecture with dynamic pricing and race-condition-free processing for transaction accuracy; contained a payment-SDK regression the same day (an estimated **\$4.8k** in bookings at risk) with a dependency pin and a CI smoke guard
- Engineer a pricing and promotions engine – tiered and peak-day rates and a competitor-aware discount matrix – applying **\$26.2k** in discounts across 1,000 redemptions
- Ship a pre-registered checkout A/B experiment across iOS and Android, with sticky per-visitor assignment and a signed, timestamped analysis plan
- Build LLM features on the Anthropic API: automated business reporting, a production log sentinel, and a read-only admin analyst that answers business questions with model-written SQL under a least-privilege Postgres role
- Review business intelligence metrics through financial analytics and occupancy forecasting to track revenue milestones

Senior QA Engineer — Embrace Pet Insurance

Jun 2025 – Dec 2025 · Remote

- Validated Salesforce CRM functionality and data models during the integration of quote creation tools across multiple brands
- Utilized Salesforce and Cosmos DB for integration testing and data synchronization validation across complex insurance systems
- Managed **400+ test cases** in Testmo to streamline defect management and ensure regression testing coverage
- Collaborated on CI/CD deployment via Azure DevOps to automate production releases and maintain platform integrity

QA Analyst — BlueinGreen (Netflix contractor)

Jun 2024 – Jun 2026 · Remote

- Validate performance of primary acquisition pages to ensure functional integrity for systems contributing to **\$1.2B** in subscription revenue (part-time from Dec 2024)
- Automate regression testing by building a Playwright framework from zero, scaling to **700+ tests** for the customer acquisition platform
- Achieve **90%** reduction in manual regression testing through strategic test automation
- Lead complex migration testing and log critical defects to ensure seamless data synchronization between vendor platforms
- Review Asset Editor features including batch operations and real-time preview functionality to validate business requirements

QA Analyst — Stake (fintech)

Mar 2022 – May 2024 · Remote

- Performed API testing with Postman and ChaiJS to validate financial transaction data synchronization and functional integrity
- Automated data queries using SQL and Python to review data accuracy and financial system reliability
- Executed integration testing with mocked data to validate business process automation and code coverage for mobile applications
- Logged integration errors and enhanced system monitoring using BigQuery to improve data accuracy by **15%**

QA Engineer — CSC Generation

Jul 2021 – May 2022 · Remote

- Reviewed and analyzed requirements with cross-functional teams to automate validation for new feature integrations
- Developed comprehensive test cases to improve regression testing accuracy and defect management efficiency

Software QA Analyst — Kemper Insurance

Apr 2021 – Jul 2021 · Remote

- Executed functional, integration, and regression testing with rigorous defect reporting and data accuracy analysis
- Validated scenario-based testing across diverse insurance applications to ensure business process integrity

Software QA Analyst — American Access Casualty

Jan 2020 – Apr 2021 · Remote

- Validated detailed test scripts and scenarios for insurance platforms, focusing on complex integration testing
- Automated test processes to validate business requirements, improving validation efficiency by **25%**
- Logged and documented defects in Jira to ensure swift resolution of data synchronization issues

Junior Application Developer — American Access Casualty

Oct 2018 – Jan 2020 · Remote

- Developed and validated web applications throughout the production lifecycle, ensuring functional integrity
- Reviewed and corrected software errors to improve data model accuracy and system reliability

PROJECTS

AI-Directed DJ Mixing System — personal project

2026

Product owner and creative director; built with Claude Code as AI engineering partner

- Designed and directed a live-performance DJ application that beat-matches a personal library of 50+ tracks across three decks. Timing is sample-accurate: in end-to-end audio tests, the two decks' kicks land about 1 ms apart.
- Integrated Anthropic's Claude as a real-time "music director". It chooses the next track, shapes each transition, and layers samples using measured key, energy and harmonic fit. An assist mode lets the DJ pick tracks while the AI handles the blend.
- Drove quality from the DJ booth. I tested in live sets, reported problems by ear (off-beat blends, harmonic clashes, controls that didn't match the sound), and turned each into a measured fix. One example: an audio audit found and corrected beat grids running half a beat out in 8 of 53 tracks.
- Shaped the experience around how a DJ works:
 - next-track suggestions that follow the track you've chosen
 - tracking of what's been played, plus a set playlist
 - a sample bank with loops
 - a history of loved and rejected transitions
 - live updates to the AI logic mid-set, without stopping the music
- Ran development as Ralph-loop-style, test-driven iteration:
 - Every requirement became failing tests first, shown to fail against the old code.
 - The AI agent then looped through fix and re-run until every test passed.
 - Independent adversarial reviews had to prove each bug they reported, and every proof became a permanent regression test.
 - Result: around 600 automated tests, plus thousands of randomized stress-test sessions against both a simulated and the real audio engine.

Skills: product direction · directing AI coding agents · Ralph-loop / test-driven development · Claude API · Python · SuperCollider · audio analysis · live DJ performance

EDUCATION

LLB, BUSINESS LAW

London Metropolitan University · 2006

FULL-STACK WEB DEVELOPMENT

Actualize Coding Bootcamp · 2018